

Final Mixed Practice Bank and Formula Recall
Part 7 of the OT Science Formula Completion Workbook

OT Science Notes

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1 Final Mixed Practice Bank and Formula Recall

This final part is designed for spaced revision. It deliberately mixes recall, computation, proof, interpretation and engineering-style applications. Use it after completing the earlier parts, or use it diagnostically to find weak areas.

1.1 Formula Recall Page

Final Formula Recall

Fill these from memory before checking earlier sections.

- | | |
|--|---|
| 1. $\nabla f =$ _____ | 14. $A'_{ij} =$ _____ |
| 2. $\nabla \cdot \mathbf{A} =$ _____ | 15. $\mathbf{g}_i =$ _____ |
| 3. $\nabla \times \mathbf{A} =$ _____ | 16. $\mathbf{g}^i \cdot \mathbf{g}_j =$ _____ |
| 4. $D_{\hat{\mathbf{u}}} f =$ _____ | 17. $A_i =$ _____ |
| 5. $\nabla^2 f =$ _____ | 18. $A^i =$ _____ |
| 6. $(\nabla \mathbf{A})_{ij} =$ _____ | 19. $ds^2 =$ _____ |
| 7. $\delta_{ij} a_j =$ _____ | 20. $dV =$ _____ |
| 8. $\varepsilon_{ijk} \varepsilon_{ilm} =$ _____ | 21. $dV_{\text{cyl}} =$ _____ |
| 9. $S_{ij} =$ _____ | 22. $dV_{\text{sph}} =$ _____ |
| 10. $W_{ij} =$ _____ | 23. $e^{i\theta} =$ _____ |
| 11. $I_1 =$ _____ | 24. $\cos(ix) =$ _____ |
| 12. $I_2 =$ _____ | 25. $\sin(ix) =$ _____ |
| 13. $I_3 =$ _____ | 26. $\cosh^2 x - \sinh^2 x =$ _____ |

1.2 Part I: Vector Calculus Computation

Final Practice I: Grad, Div, Curl and Directional Derivatives

1. Compute ∇f for $f = x^3 y^2 + z \cos x$.
2. Compute ∇f for $f = \ln(x^2 + y^2 + z^2)$, away from the origin.
3. Compute $\nabla^2 f$ for $f = x^2 + y^2 + z^2$.
4. Compute $\nabla^2 f$ for $f = 1/r$, where $r = \sqrt{x^2 + y^2 + z^2}$, away from the origin.
5. Compute $\nabla \cdot \mathbf{A}$ for $\mathbf{A} = x^2 \hat{\mathbf{i}} + y^2 \hat{\mathbf{j}} + z^2 \hat{\mathbf{k}}$.
6. Compute $\nabla \cdot \mathbf{A}$ for $\mathbf{A} = yz \hat{\mathbf{i}} + xz \hat{\mathbf{j}} + xy \hat{\mathbf{k}}$.
7. Compute $\nabla \times \mathbf{A}$ for $\mathbf{A} = xy \hat{\mathbf{i}} + yz \hat{\mathbf{j}} + zx \hat{\mathbf{k}}$.
8. Compute $\nabla \times \mathbf{A}$ for $\mathbf{A} = (-y/(x^2 + y^2)) \hat{\mathbf{i}} + (x/(x^2 + y^2)) \hat{\mathbf{j}}$, away from the z -axis.
9. For $f = x^2 y + yz^2$, compute the directional derivative at $(1, 1, 2)$ in direction $(2, -1, 2)$.
10. Find the direction of steepest increase of $f = xy + yz + zx$ at $(1, 2, 3)$.

11. Compute $\nabla \mathbf{A}$ for $\mathbf{A} = x^2 \hat{\mathbf{i}} + xy \hat{\mathbf{j}} + xz \hat{\mathbf{k}}$.
12. For the vector field in Question 11, find the symmetric and skew-symmetric parts of $\nabla \mathbf{A}$.
13. Verify $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ for $\mathbf{A} = z \hat{\mathbf{i}} + x \hat{\mathbf{j}} + y \hat{\mathbf{k}}$.
14. Verify $\nabla \times (\nabla f) = 0$ for $f = xyz + x^2 + y^2$.
15. Evaluate $\nabla \times (\nabla \times \mathbf{A})$ directly and using the identity for $\mathbf{A} = x^2 \hat{\mathbf{i}} + y^2 \hat{\mathbf{j}} + z^2 \hat{\mathbf{k}}$.

1.3 Part II: Coordinate Systems and Differential Operators

Final Practice II: Coordinates, Scale Factors and Operators

1. Convert $(x, y, z) = (-1, \sqrt{3}, 2)$ to cylindrical coordinates using atan2.
2. Convert $(x, y, z) = (1, 1, \sqrt{2})$ to spherical coordinates.
3. Convert $(\rho, \phi, z) = (3, 5\pi/6, -2)$ to Cartesian coordinates.
4. Convert $(r, \theta, \phi) = (2, \pi/3, \pi/4)$ to Cartesian coordinates.
5. Derive the cylindrical basis vectors $\hat{\mathbf{e}}_\rho$ and $\hat{\boldsymbol{\phi}}$ in terms of $\hat{\mathbf{i}}, \hat{\mathbf{j}}$.
6. Derive the spherical basis vectors $\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi$ in terms of $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$.
7. Derive h_ρ, h_ϕ, h_z from the cylindrical position vector.
8. Derive h_r, h_θ, h_ϕ from the spherical position vector.
9. Write $[g_{ij}]$ and $[g^{ij}]$ for cylindrical coordinates.
10. Write $[g_{ij}]$ and $[g^{ij}]$ for spherical coordinates.
11. Compute ∇f in cylindrical coordinates for $f = \rho z \cos \phi$.
12. Compute ∇f in spherical coordinates for $f = r^3 \sin \theta \cos \phi$.
13. Compute $\nabla \cdot \mathbf{A}$ in cylindrical coordinates for $\mathbf{A} = \rho \hat{\mathbf{e}}_\rho + \rho^2 \hat{\boldsymbol{\phi}} + z \hat{\mathbf{k}}$.
14. Compute $\nabla \cdot \mathbf{A}$ in spherical coordinates for $\mathbf{A} = r \hat{\mathbf{e}}_r + \sin \theta \hat{\mathbf{e}}_\theta + \cos \phi \hat{\mathbf{e}}_\phi$.
15. Compute $\nabla \times \mathbf{A}$ in cylindrical coordinates for $\mathbf{A} = 0 \hat{\mathbf{e}}_\rho + \rho \hat{\boldsymbol{\phi}} + z \hat{\mathbf{k}}$.
16. Compute $\nabla \times \mathbf{A}$ in spherical coordinates for $\mathbf{A} = 0 \hat{\mathbf{e}}_r + 0 \hat{\mathbf{e}}_\theta + r \sin \theta \hat{\mathbf{e}}_\phi$.
17. Derive the cylindrical scalar Laplacian from the general orthogonal coordinate formula.
18. Derive the spherical scalar Laplacian from the general curvilinear formula.

1.4 Part III: Index Notation and Tensor Identities

Final Practice III: Index Notation, Delta and Levi-Civita

1. Expand $A_{ij}B_{jk}$ for the component $(i, k) = (1, 2)$.
2. Simplify $\delta_{ij}\delta_{jk}a_k$.
3. Simplify $\delta_{ij}A_{ik}B_{jk}$.
4. Simplify $\varepsilon_{ijk}\varepsilon_{ilm}a_jb_kc_ld_m$.
5. Show that $\varepsilon_{ijk}a_jb_k$ gives the cross product.
6. Prove $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$.
7. Prove $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.
8. Prove the vector triple product identity using ε_{ijk} .
9. Prove $\nabla \times (\nabla f) = 0$ using indices.
10. Prove $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ using indices.
11. Prove the curl-curl identity using indices.
12. Compute $\partial_i(x_jx_j)$.
13. Compute $\partial_j(x_ix_j)$ in three dimensions.
14. Compute $\partial_i(r^n)$, where $r = (x_jx_j)^{1/2}$.
15. Compute $\partial_i(x_i/r^3)$ for $r \neq 0$.

1.5 Part IV: Tensor Decompositions and Invariants

Final Practice IV: Symmetry, Skew-Symmetry and Invariants

1. Decompose $A = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 4 & 5 \\ 1 & -2 & 6 \end{pmatrix}$ into symmetric and skew parts.
2. Check whether $A = \begin{pmatrix} 0 & -2 & 1 \\ 2 & 0 & -3 \\ -1 & 3 & 0 \end{pmatrix}$ is skew-symmetric.
3. Compute I_1, I_2, I_3 for $A = \text{diag}(2, 4, 6)$.
4. Compute I_1, I_2, I_3 for $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.
5. Find the characteristic polynomial of $A = \text{diag}(a, b, c)$.
6. Find the deviatoric part of $A = \text{diag}(5, 2, 2)$.
7. Find the hydrostatic and deviatoric parts of $\sigma = \text{diag}(120, 60, 30)$.
8. Prove that the deviatoric tensor has zero trace.

9. Prove that $x_i W_{ij} x_j = 0$ for skew-symmetric W .
10. Show that $x_i A_{ij} x_j$ depends only on the symmetric part of A .
11. For $\mathbf{v} = (-\omega y, \omega x, 0)$, compute the velocity gradient and split it into symmetric and skew parts.
12. For $\mathbf{v} = (ax, by, cz)$, compute the invariants of $\nabla \mathbf{v}$.

1.6 Part V: Covariant and Contravariant Components

Final Practice V: Metrics, Raising and Lowering

1. Explain the difference between $\mathbf{g}_i$ and $\mathbf{g}^i$.
2. Prove $\mathbf{g}^i \cdot \mathbf{g}_j = \delta_j^i$ for an orthogonal coordinate system.
3. In cylindrical coordinates, compute $\mathbf{g}_\rho, \mathbf{g}_\phi, \mathbf{g}_z$.
4. In cylindrical coordinates, compute $\mathbf{g}^\rho, \mathbf{g}^\phi, \mathbf{g}^z$.
5. In spherical coordinates, compute $\mathbf{g}_r, \mathbf{g}_\theta, \mathbf{g}_\phi$.
6. In spherical coordinates, compute $\mathbf{g}^r, \mathbf{g}^\theta, \mathbf{g}^\phi$.
7. Given cylindrical physical components $(A_\rho, A_\phi, A_z) = (3, 4, 5)$, find A^i and A_i .
8. Given spherical physical components $(A_r, A_\theta, A_\phi) = (2, 3, 4)$, find A^i and A_i .
9. Prove $\mathbf{A} \cdot \mathbf{B} = g_{ij} A^i B^j$.
10. Derive $\nabla^2 f = \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} g^{ij} \partial_j f)$ for the cylindrical case.

1.7 Part VI: Complex Numbers and Hyperbolic Functions

Final Practice VI: Complex and Hyperbolic Problems

1. Write $z = -1 + i$ in polar and exponential form.
2. Write $z = -\sqrt{3} - i$ in polar and exponential form.
3. Compute $(\sqrt{3} + i)^6$ using polar form.
4. Find all roots of $z^4 = 16$.
5. Find all roots of $z^3 = -8$.
6. Solve $z^5 = 32e^{i\pi/2}$.
7. Prove Euler's formula from the power series of e^x , $\sin x$, and $\cos x$.
8. Derive De Moivre's theorem.
9. Express $\cos(x + iy)$ in real and imaginary parts.
10. Express $\sin(x + iy)$ in real and imaginary parts.
11. Prove $\cos(ix) = \cosh x$.
12. Prove $\sin(ix) = i \sinh x$.

13. Prove $\tanh(ix) = i \tan x$.
14. Derive $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$.
15. Derive $\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$.

1.8 Part VII: Mixed Challenge Problems

Final Practice VII: Mixed Challenge Set

1. A scalar field is given by $T = 1/r$. Compute ∇T , $\nabla^2 T$ for $r \neq 0$, and interpret the singularity at the origin qualitatively.
2. A velocity field is $\mathbf{v} = a\rho\hat{\mathbf{e}}_\rho + bz\hat{\mathbf{k}}$ in cylindrical coordinates. Find $\nabla \cdot \mathbf{v}$. For what relation between a and b is the flow incompressible?
3. A purely azimuthal field is $\mathbf{A} = A_\phi(r, \theta)\hat{\mathbf{e}}_\phi$ in spherical coordinates. Write the general formula for its curl.
4. A tensor A has eigenvalues 1, 2, 5. Find I_1, I_2, I_3 and write its characteristic polynomial.
5. For $A_{ij} = x_i x_j$, compute $\text{tr}(A)$, $A_{ij}A_{ij}$, and $\partial_j A_{ij}$.
6. Use $\varepsilon_{ijk}\varepsilon_{ilm}$ to derive Lagrange's identity $|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2$.
7. Show that $\nabla \cdot (f\mathbf{A} \times \mathbf{B})$ can be expanded using product identities.
8. In spherical coordinates, compute ∇f and $\nabla^2 f$ for $f = f(r)$, a radial-only scalar field.
9. In cylindrical coordinates, compute ∇f and $\nabla^2 f$ for $f = f(\rho)$, an axisymmetric radial field.
10. Explain why a formula that is true in Cartesian components may need correction terms in curvilinear coordinates if components are not physical components.
11. A complex number is multiplied by $2e^{i\pi/3}$. Describe the geometric action.
12. Solve $e^{2z} = 1$ for complex z .
13. Show that $\cosh x$ and $\sinh x$ parametrise the unit hyperbola $X^2 - Y^2 = 1$.
14. Compare $\cos^2 x + \sin^2 x = 1$ with $\cosh^2 x - \sinh^2 x = 1$. Explain the sign difference using exponentials.
15. Build a one-page personal formula sheet from memory covering all topics in this workbook.